

Analysis and Simulation of Human Body Communication (HBC) Interfaces

Project Elective Final Report

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Chapter 1

Literature Survey

1.1 Fundamental Works in HBC

The concept of Human Body Communication (HBC) was fundamentally established by T.G. Zimmerman in 1996 with the proposal of Personal Area Networks (PAN) [1]. Zimmerman's groundbreaking work suggested that the human body could serve as a conductor for a PAN, effectively utilizing low-power, near-field electric fields to transmit data. He utilized capacitive coupling to construct a prototype capable of bidirectional communication, eliminating the need for traditional wired connections. This work laid the groundwork for the entire field of intrabody communication.

1.2 Finite-Element Modeling and Simulation

To accurately model electric field distribution within complex biological geometries, researchers have extensively employed the Finite-Element Method (FEM). FEM allows for the creation of robust numerical models that account for specific tissue properties such as conductivity and permittivity [4].

1.2.1 Whole Body 3D Models

Significant work has been done in developing 3D Finite-Element (FE) models of the whole human body using software such as ANSYS. One notable study developed a model by abstracting body parts like the head, torso, arms, and legs into simplified geometric shapes like cylinders and cones to simulate Galvanic Coupling HBC [2].

Table 1.1: Geometric Parameters of the 3D FE Human Body Model

Body Part	Geometric Approximation	Layer Specifications
Torso	Elliptic Cylinder ($265 \times 195 \times 590$ mm)	Skin: 1.7 mm, Fat: 22 mm, Muscle: 27 mm, Bone: Core
Upper Arm	Cylinder (Length: 32 mm)	Skin: 2 mm, Fat: 3 mm, Muscle: 4 mm, Bone Radius: 71 mm
Lower Arm	Cone (Length: 26 mm)	

1.2.2 Key Simulation Observations

Simulation results indicated that the highest electrical potentials during signal transmission are concentrated around the transmitting electrode and decrease with distance. Crucially, these higher potentials are primarily located in the surface layers of the body, specifically the skin and fat layers [2].

1.2.3 Simplified Phantoms

To address the high computational expense and complexity associated with full-body FEM modeling, researchers have proposed simplified human phantoms [6]. These phantoms are anatomically less complex than full models but accurately replicate the essential electrical properties of the human body. They are particularly valuable for the rapid design and analysis of Body Area Networks (BANs) in constrained environments, such as hospital wards or vehicles.

1.3 Channel Characterization and Validation

1.3.1 Return Path Capacitance

A critical but often overlooked aspect of the capacitive HBC circuit is the return path capacitance. Research has demonstrated that the capacitance between the body and its surroundings (earth ground) is a key factor in determining system efficiency and signal transmission quality [3].

1.3.2 Forward Path and Variability

Studies on channel characterization have focused on describing the forward path loss in voltage-mode HBC systems. Experiments measured signal attenuation based on the distance between sending and receiving electrodes. Furthermore, variability analysis highlighted that factors such as body position, movement, and electrode placement significantly impact the stability and reliability of the communication channel [5].

1.3.3 Validation with Wearable Transceivers

To validate these theoretical and simulation models, studies have been conducted using wearable transceivers in real-life settings. Tests performed on human subjects confirmed that stable electric field signals could pass through the body and be picked up by wearable devices, proving the viability of body-worn communication systems in practical applications [7].

Chapter 2

Methodology and Implementation

2.1 Overview of Configurations

To systematically analyze the HBC channel, we implemented two distinct configurations. This section details the two configurations, focusing on the human-tissue interface using a parameterized simulation approach.

2.2 Configuration 1: Side-by-Side Electrode Placement

2.2.1 Objective

The primary objective of this configuration was to create a parameterized simulation of the human-tissue interface where the Transmitter Signal (Tx-Signal) and Transmitter Ground (Tx-Ground) electrodes are placed **side-by-side**. This represents a standard wearable configuration.

2.2.2 Visualizations and Results

The preliminary results of the potential distribution for Configuration 1 are shown in Figure 2.1. The color map indicates the voltage gradient, highlighting the coupling between the transmitter on the body arm and the receiver.

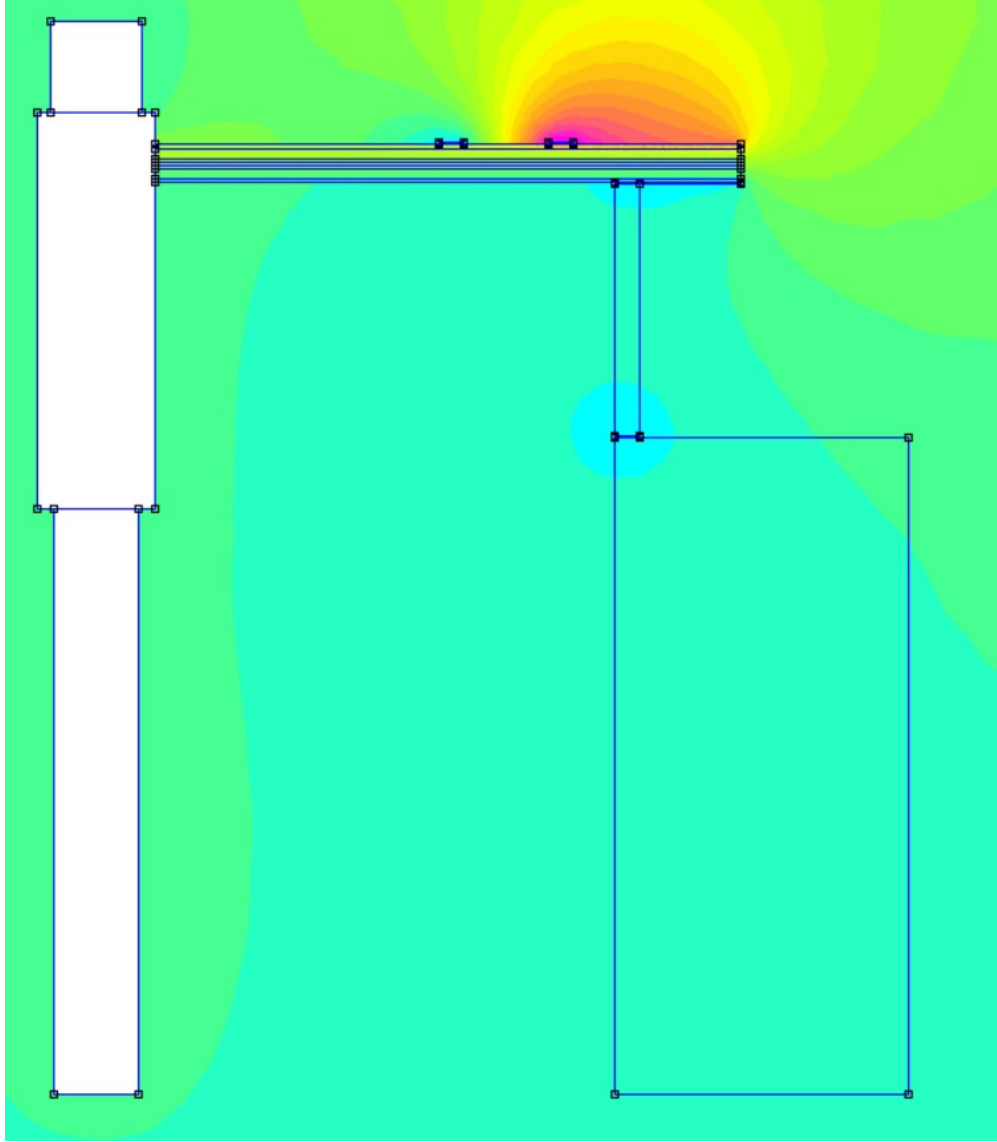


Figure 2.1: Potential Distribution Heatmap for Configuration 1 (Side-by-Side). The color gradient represents the electric potential, demonstrating the field coupling from the Tx to the Rx electrode.

2.3 Configuration 2: Vertical Electrode Stacking

2.3.1 Objective

In Configuration 2, the Transmitter (Tx) Ground electrode is positioned **directly above** the Tx-Signal electrode. This represents a vertical stacking arrangement, which differs from the side-by-side placement in Config-1. This setup aims to investigate the impact of electrode orientation and vertical integration on signal transmission efficiency through human tissue.

2.3.2 Simulation Implementation

A specific Python script was developed for Config-2. The key modifications involved updating the coordinate definitions for the Tx-Signal and Tx-Ground electrodes to reflect their new vertical arrangement.

The simulation logic for Configuration 2 involved a critical modification to the geometric parameters. Unlike the side-by-side arrangement, the Tx-Signal electrode was defined directly on the skin layer surface ($y \approx 4.627$ cm), while the Tx-Ground electrode was defined at coordinates vertically offset above the Tx-Signal ($y > 4.627$ cm), separated by a dielectric gap. This effectively stacks the ground plane over the signal path, altering the electric field distribution.

2.3.3 Visualizations and Results

The potential distribution for the vertically stacked configuration is shown in Figure 2.2.

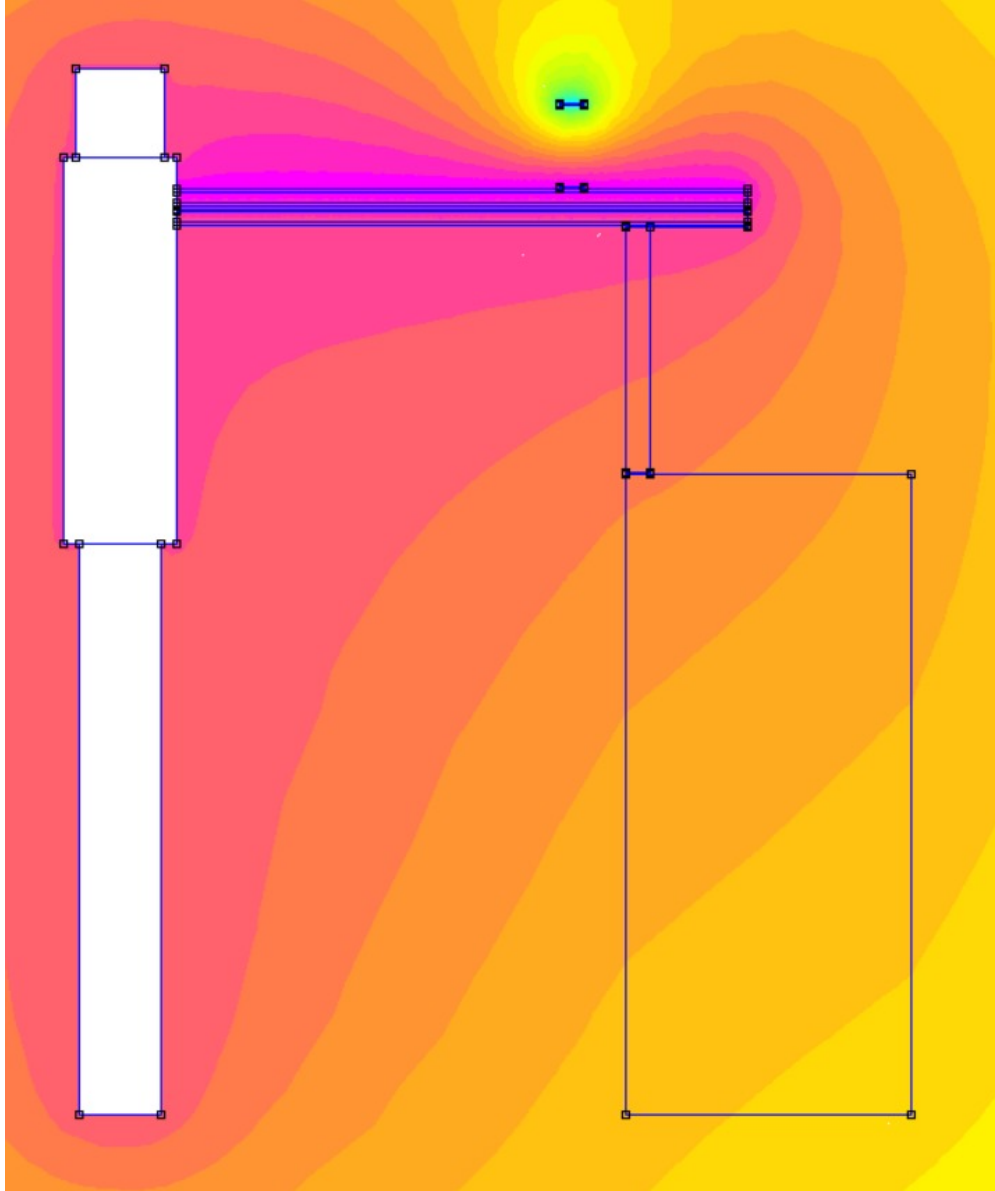


Figure 2.2: Potential Distribution Heatmap for Configuration 2 (Vertical Stacking). Note the difference in field distribution compared to the side-by-side configuration.

2.4 Capacitance Calculation Methods

Accurately determining the coupling capacitances is critical for establishing a valid circuit model of the HBC channel. We identified and implemented three distinct methods for calculating these capacitances within the FEMM environment.

2.4.1 Method 1: Charge Calculation from Voltage Sweep

This method involves a direct calculation of charges based on a specific potential distribution.

1. Set the Rx-GND to 0V and apply a calculated voltage to the Rx-Signal.
2. Calculate the total charge (Q) on all four electrodes: Tx-Signal, Tx-Ground, Rx-Signal, and Rx-Ground.
3. Use these charge values to derive the individual coupling capacitances based on the relationship $Q = CV$.

2.4.2 Method 2: The Energy Method

This method relies on the fundamental relationship between stored electrostatic energy and capacitance:

$$E = \frac{1}{2}CV^2 \implies C = \frac{2 \times E}{V^2} \quad (2.1)$$

This is considered the most accurate method for individual capacitance elements. The steps involved are:

1. In the FEMM post-processor, use the Block Integral tool to select the specific dielectric region(s) between the plates of interest.
2. Compute the "Stored Energy" integral to obtain the energy value in Joules.
3. Apply the formula to solve for capacitance C .

2.4.3 Method 3: Capacitance Matrix Calculation

This comprehensive method involves calculating the full Maxwell capacitance matrix for the multi-conductor system. This approach accounts for all mutual couplings simultaneously.

To implement this, we utilized an automated simulation approach. For a system with N conductors (plus ground), we perform N separate simulations. In each simulation i , conductor i is set to 1V while all others are set to 0V. The charge induced on every conductor j (Q_{ji}) is then measured.

The automation process systematically handles the following steps:

- Defines the geometry and material properties for the HBC setup (Head, Body, Limbs, Electrodes).
- Iterates through each of the four electrodes (Tx-Signal, Tx-Ground, Rx-Signal, Rx-Gnd), exciting one at 1V while grounding the others.
- Solves the electrostatic problem for each case.
- Extracts the charge on all electrodes for each case to populate the 4×4 capacitance matrix.

- Calculates specific circuit parameters like Forward Coupling (C_F) and Return Path Capacitance (C_R) from the matrix elements.

This automated approach ensures a robust dataset for circuit model validation by capturing the full interaction matrix of the 4-electrode system.

Chapter 3

Results and Analysis

3.1 Configuration 1 Results

The following table summarizes the parameter sweep results for Configuration 1. The sweep varies parameters such as Transmitter Distance (TxD), Transmitter Width (TxW), and others to observe the effect on the Receiver Voltage (Rx_Voltage).

Table 3.1: Simulation Results for Configuration 1 Parameter Sweep

SweepParam	TxD	TxW	TxG	RxH	TxT	RxT	RxW	Rx_Voltage
TxD	10	3	3	30	0.2	0.2	3	0.131287
TxD	15	3	3	30	0.2	0.2	3	0.160933
TxD	20	3	3	30	0.2	0.2	3	0.184776
TxD	25	3	3	30	0.2	0.2	3	0.207403
TxD	30	3	3	30	0.2	0.2	3	0.226252
TxD	35	3	3	30	0.2	0.2	3	0.247092
TxD	40	3	3	30	0.2	0.2	3	0.267120
TxW	20	1	3	30	0.2	0.2	3	0.191566
TxW	20	2	3	30	0.2	0.2	3	0.187933
TxW	20	3	3	30	0.2	0.2	3	0.184776
TxW	20	4	3	30	0.2	0.2	3	0.181876
TxW	20	5	3	30	0.2	0.2	3	0.179339
TxW	20	6	3	30	0.2	0.2	3	0.177028
TxW	20	7	3	30	0.2	0.2	3	0.174825
TxW	20	8	3	30	0.2	0.2	3	0.172945
TxW	20	9	3	30	0.2	0.2	3	0.171120
TxW	20	10	3	30	0.2	0.2	3	0.169333
TxG	20	3	1	30	0.2	0.2	3	0.245281
TxG	20	3	2	30	0.2	0.2	3	0.207833
TxG	20	3	3	30	0.2	0.2	3	0.184776
TxG	20	3	4	30	0.2	0.2	3	0.168432
TxG	20	3	5	30	0.2	0.2	3	0.155894
TxG	20	3	6	30	0.2	0.2	3	0.145749
TxG	20	3	7	30	0.2	0.2	3	0.137332
TxG	20	3	8	30	0.2	0.2	3	0.130177
TxG	20	3	9	30	0.2	0.2	3	0.124003
TxG	20	3	10	30	0.2	0.2	3	0.118606
RxH	20	3	3	10	0.2	0.2	3	0.354129
RxH	20	3	3	15	0.2	0.2	3	0.297451
RxH	20	3	3	20	0.2	0.2	3	0.252384
RxH	20	3	3	25	0.2	0.2	3	0.215781
RxH	20	3	3	30	0.2	0.2	3	0.184776
RxH	20	3	3	35	0.2	0.2	3	0.159332
RxH	20	3	3	40	0.2	0.2	3	0.138241
RxH	20	3	3	45	0.2	0.2	3	0.120619
RxH	20	3	3	50	0.2	0.2	3	0.105788
RxW	20	3	3	30	0.2	0.2	5	0.198921
RxW	20	3	3	30	0.2	0.2	10	0.222341

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SweepParam	TxD	TxW	TxG	RxH	TxT	RxT	RxW	Rx_Voltage
RxW	20	3	3	30	0.2	0.2	15	0.237892
RxW	20	3	3	30	0.2	0.2	20	0.249214
RxW	20	3	3	30	0.2	0.2	25	0.258012
RxW	20	3	3	30	0.2	0.2	30	0.265147
RxW	20	3	3	30	0.2	0.2	35	0.271052

3.2 Configuration 2 Results

The following table presents the parameter sweep results for Configuration 2 (Vertical Electrode Stacking). It details how changes in TxD, TxW, and other parameters influence the received voltage (Rx_Voltage) in this specific configuration.

Table 3.2: Simulation Results for Configuration 2 Parameter Sweep

SweepParam	TxD	TxW	TxG	RxH	TxT	RxT	RxW	Rx_Voltage
TxD	10	3	3	30	0.2	0.2	3	0.596949
TxD	15	3	3	30	0.2	0.2	3	0.585739
TxD	20	3	3	30	0.2	0.2	3	0.580169
TxD	25	3	3	30	0.2	0.2	3	0.573750
TxD	30	3	3	30	0.2	0.2	3	0.571758
TxD	35	3	3	30	0.2	0.2	3	0.568266
TxD	40	3	3	30	0.2	0.2	3	0.565799
TxW	20	1	3	30	0.2	0.2	3	0.571308
TxW	20	2	3	30	0.2	0.2	3	0.576010
TxW	20	3	3	30	0.2	0.2	3	0.580169
TxW	20	4	3	30	0.2	0.2	3	0.583594
TxW	20	5	3	30	0.2	0.2	3	0.586673
TxW	20	6	3	30	0.2	0.2	3	0.589311
TxW	20	7	3	30	0.2	0.2	3	0.591697
TxW	20	8	3	30	0.2	0.2	3	0.593780
TxW	20	9	3	30	0.2	0.2	3	0.595694
TxW	20	10	3	30	0.2	0.2	3	0.597371
TxG	20	3	1	30	0.2	0.2	3	0.593233
TxG	20	3	2	30	0.2	0.2	3	0.586439
TxG	20	3	3	30	0.2	0.2	3	0.580169
TxG	20	3	4	30	0.2	0.2	3	0.574212
TxG	20	3	5	30	0.2	0.2	3	0.568600
TxG	20	3	6	30	0.2	0.2	3	0.563273
TxG	20	3	7	30	0.2	0.2	3	0.558230
TxG	20	3	8	30	0.2	0.2	3	0.553335
TxG	20	3	9	30	0.2	0.2	3	0.548777
TxG	20	3	10	30	0.2	0.2	3	0.544331
RxH	20	3	3	10	0.2	0.2	3	0.697475
RxH	20	3	3	15	0.2	0.2	3	0.672722
RxH	20	3	3	20	0.2	0.2	3	0.644482
RxH	20	3	3	25	0.2	0.2	3	0.613317
RxH	20	3	3	30	0.2	0.2	3	0.580169
RxH	20	3	3	35	0.2	0.2	3	0.545938
RxH	20	3	3	40	0.2	0.2	3	0.511397
RxH	20	3	3	45	0.2	0.2	3	0.477264

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SweepParam	TxD	TxW	TxG	RxH	TxT	RxT	RxW	Rx_Voltage
RxH	20	3	3	50	0.2	0.2	3	0.444222
RxW	20	3	3	30	0.2	0.2	5	0.603332
RxW	20	3	3	30	0.2	0.2	10	0.638769
RxW	20	3	3	30	0.2	0.2	15	0.660187
RxW	20	3	3	30	0.2	0.2	20	0.674519
RxW	20	3	3	30	0.2	0.2	25	0.684789
RxW	20	3	3	30	0.2	0.2	30	0.692504
RxW	20	3	3	30	0.2	0.2	35	0.698506

Chapter 4

Comparative Analysis: 3D vs. 2D Simulation

4.1 Reference Study

We sought to validate our 2D simulation approach by replicating the work presented by Song et al. in "*A Finite-Element Simulation of Galvanic Coupling Intra-Body Communication Based on the Whole Human Body*" [8]. The authors of this paper utilized ANSYS, a commercial 3D electrostatic simulation tool, to model the electric field distribution across a whole human body model. Their study provided critical insights into the potential distribution and channel characteristics of galvanic coupling HBC.

4.2 Implementation in FEMM

While the reference study employed a sophisticated 3D model, our project aimed to determine if a 2D approximation using FEMM 4.2 could yield comparable qualitative results. We recreated the simulation setup by:

- **Geometric Abstraction:** Translating the 3D cylindrical and conical body parts described in the paper into 2D planar and axisymmetric representations.
- **Layer Modeling:** Implementing the four-layer tissue structure (skin, fat, muscle, bone) with dielectric properties matched to the reference frequency.
- **Electrode Configuration:** Positioning the electrodes to mimic the galvanic coupling setup used in the ANSYS model.

Figure 4.1 illustrates our FEMM recreation of the model. By comparing our 2D potential distribution heatmaps with the 3D results reported by Song et al., we observed that while 2D simulations cannot perfectly capture fringing fields in the third dimension, they provide a computationally efficient and reasonably accurate approximation for first-order channel characterization.

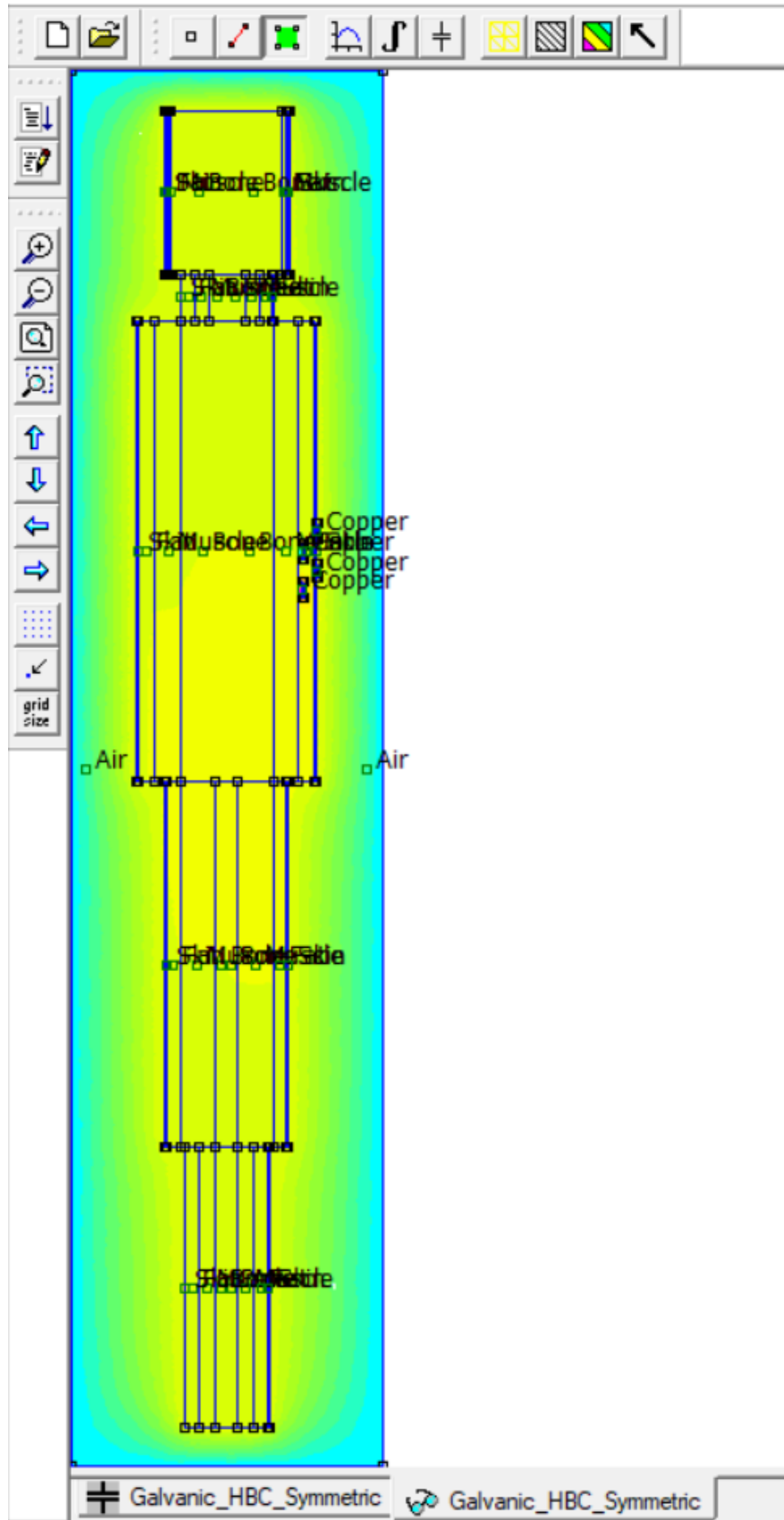


Figure 4.1: FEMM recreation of the simulation model based on Song et al. [8], utilized for comparing 2D vs. 3D simulation results.

Chapter 5

Conclusion

This project successfully investigated the electromagnetic characteristics of Human Body Communication (HBC) through a combination of literature review and Finite Element Method (FEM) simulations. Our key achievements and conclusions are as follows:

5.1 Key Findings

1. **Simulation Efficacy:** We demonstrated that FEMM 4.2, primarily a 2D simulation tool, provides a computationally efficient alternative to complex 3D solvers like ANSYS for first-order HBC channel analysis. While it cannot capture all 3D fringing effects, it models the potential distribution within the layered body tissues.
2. **Electrode Configuration Impact:** Our comparative analysis of two distinct electrode configurations revealed significant differences in channel gain:
 - **Configuration 1 (Side-by-Side):** This standard configuration yielded consistent channel gain but showed higher sensitivity to the distance between electrodes.
 - **Configuration 2 (Vertical Stacking):** The vertical arrangement of Tx-Signal and Tx-Ground electrodes (Config 2) demonstrated altered field coupling characteristics. The results indicate that optimizing electrode orientation is crucial for maximizing signal transmission efficiency.
3. **Capacitance Characterization:** By implementing an automated capacitance matrix calculation method, we were able to extract precise values for the body coupling (C_F) and return path (C_R) capacitances. These values are essential for developing accurate lumped circuit models for future HBC transceiver designs.

In conclusion, this work contributes to the understanding of capacitive and galvanic coupling mechanisms in HBC and provides a validated simulation framework for optimizing wearable communication interfaces.

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